



Foundational Model for Rover Autonomy

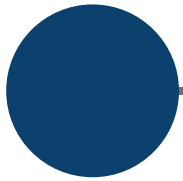
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MSc Astronautics and Space Engineering

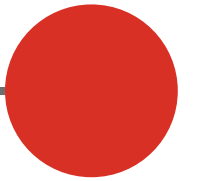
A rover cannot ask Earth what it is looking at

3 to 22 minutes



Earth

one way, at the speed of light

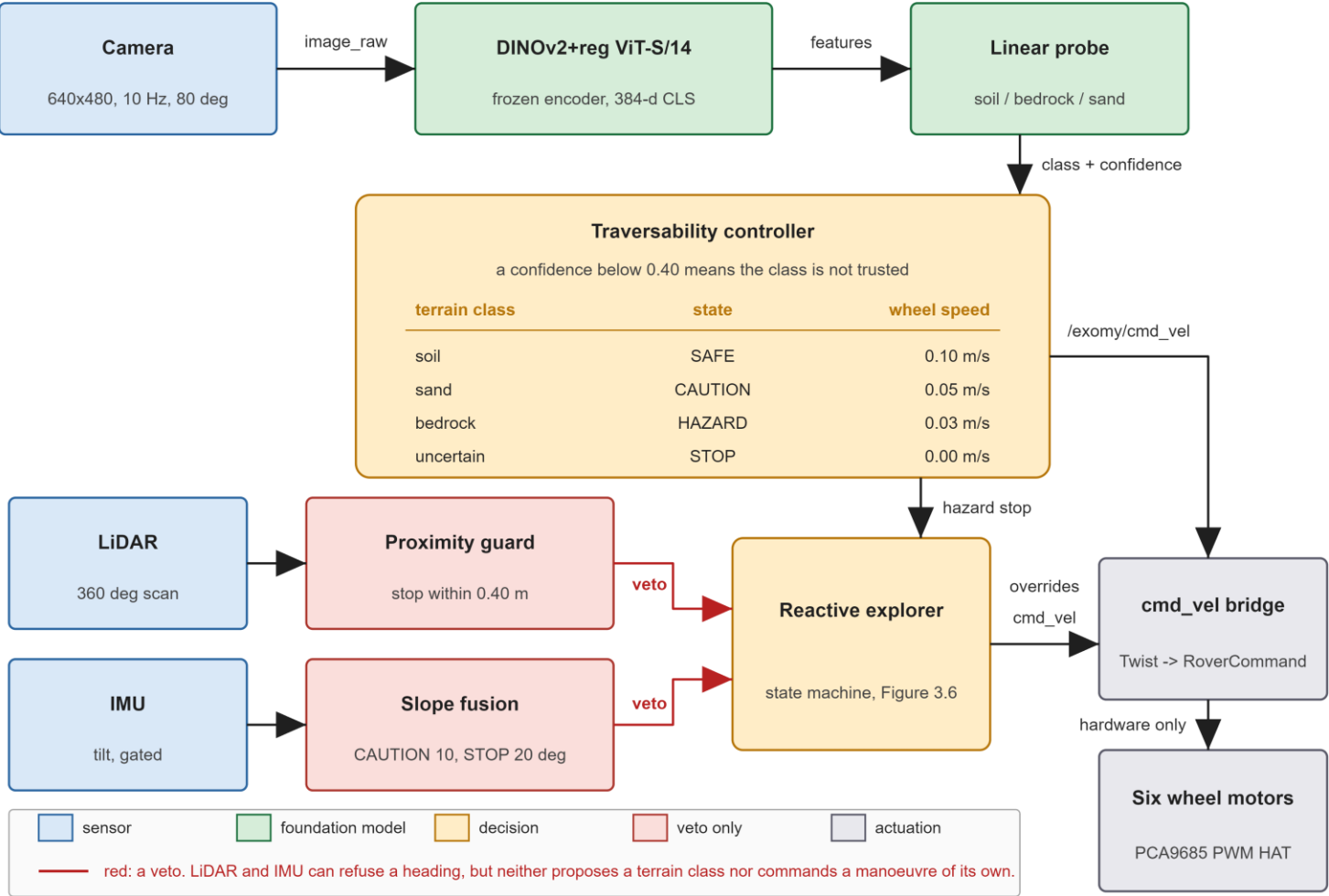


Mars

One-way light time, Earth to Mars, at closest and furthest approach.

Today's answer needs recalibration for every mission

Perception to motion: the foundation model decides where the rover may drive



Stereo geometry measures shape, not what the ground is made of.

Traversability rules are hand-tuned per mission and per site.

Learned alternatives need labelled Mars data and a GPU to train on.

Can a frozen foundation model replace that pipeline?

1

Benchmark 22 model variants across six pretraining paradigms on AI4Mars, every encoder frozen.

2

Deploy the strongest encoder that fits a Raspberry Pi 4, inside a ROS2 stack.

3

Turn class and confidence into a conservative speed policy, in Gazebo and on a real ExoMy rover.

Research question, then the three objectives that answer it.

Four questions this talk answers

1

Which pretraining wins?

2

Does it fit on the rover?

3

Is it safe when it is wrong?

4

Where does it break?



PART 1

Background

What a foundation model is, what AI4Mars contains, and why the encoder is never touched.

A foundation model is a reusable encoder, not a Mars model

1

Trained once on very large unlabelled image collections, then used unchanged for tasks it never saw.

2

The output is a feature vector per image, not a class. A small head decides what the features mean.

3

Nothing in this pipeline was trained on Mars imagery before the probe was fitted.

The encoder is the expensive part, and it is exactly the part this work does not train.

Six pretraining paradigms, six different things to learn

1

Self-distillation, DINOv2 and DINOv3: match your own output across two views of one image.

2

Contrastive and language-supervised, CLIP and SigLIP2: match an image to its caption.

3

Masked and autoregressive, EVA-02 and AIMv2, plus segmentation and depth specialists.

The comparison is between what each objective forces the encoder to encode.

AI4Mars: real rover imagery, labelled by volunteers

1

Curiosity and Perseverance navigation and mast camera frames, four terrain classes.

2

16,064 training images. A separate gold test set of 287 images, agreed by expert reviewers.

3

The class distribution is severely uneven, which turns out to matter more than anything else here.

Every accuracy in this deck is measured on the 287-image gold set.

Where this thesis sits in the autonomy stack

Layer	Function	This thesis	Verified on
L1	Terrain semantics from the camera	Core contribution, 90.24% deployed, 94.43% ceiling	Data, sim, hardware
L2	Reactive safety: stop, slow or go	Implemented and validated, 10 of 10 safe rounds	Sim, hardware
L3	Traversability as a continuous cost	Partial, discrete gating validated	Sim, hardware
L4	Localisation: where is the rover	Partial, 0.18 m SLAM error given odometry	Simulation only
L5	Path planning: how to reach a goal	Not attempted in full, A* reached a goal	Simulation only
L6	Mission autonomy: what to do next	Post-thesis, small slivers work live	Simulation only

The contribution is L1 and L2 on low-cost hardware, with partial progress on L3.



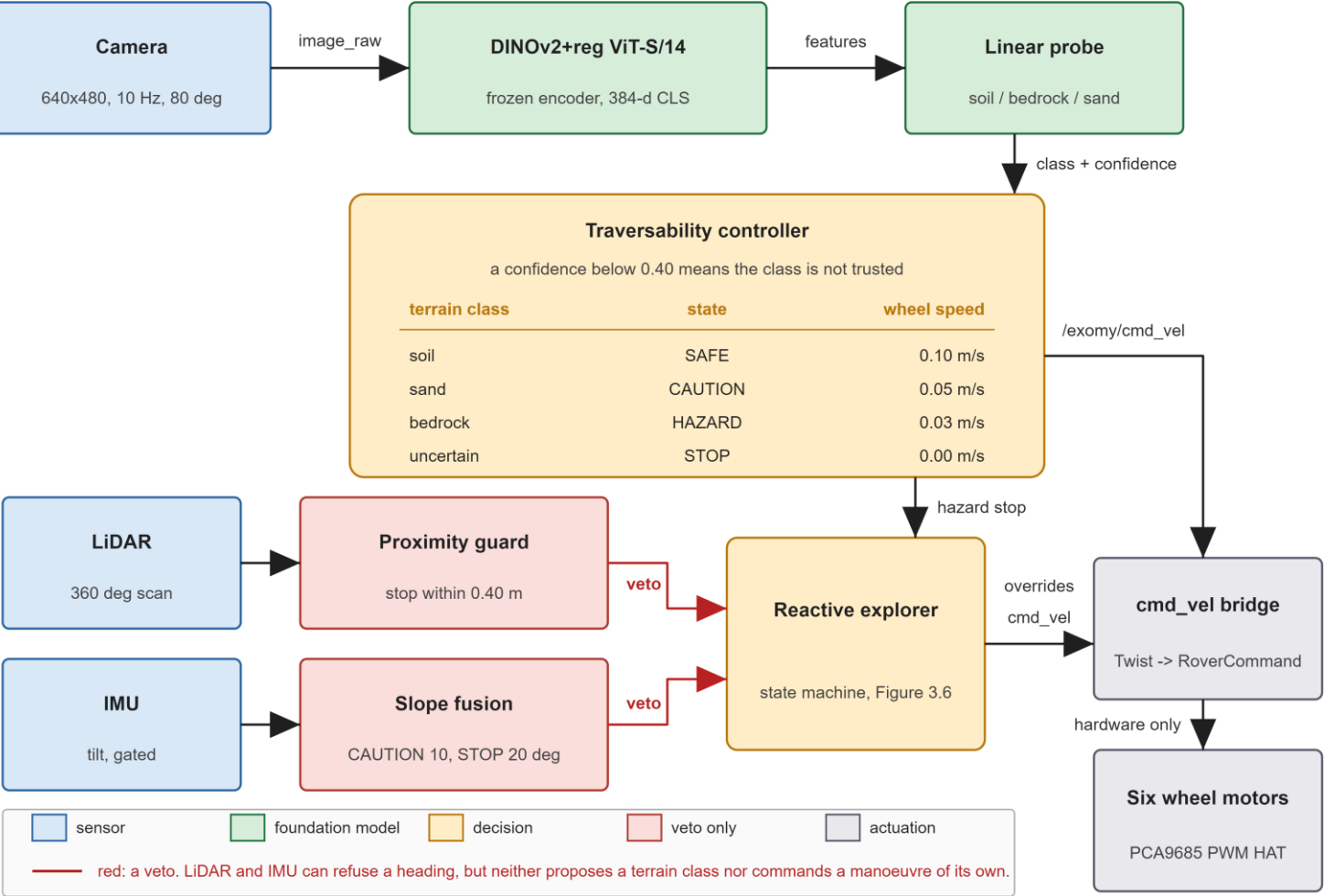
PART 2

Method

One frozen encoder, one linear probe, and a rover that has to act on the answer.

Nothing is retrained. Only a linear probe is fitted

Perception to motion: the foundation model decides where the rover may drive



A frozen DINOv2 backbone produces the features. Only a small linear probe is fitted on top.

Class and confidence set the wheel speed. Below 0.40 confidence becomes a stop.

LiDAR and IMU are vetoes only. Neither ever proposes a terrain class.

22 variants, six paradigms, one frozen protocol

Evaluation protocol	Value
Model variants compared	22
Pretraining paradigms	6
Test set	AI4Mars gold, 287 images
Labels per class, headline setting	1,000
Encoder weights updated	none
Supervised ceiling, DeepLabv3+	96.67%

Every encoder frozen, the same probe, the same 287-image gold test set.



QUESTION 1

Which pretraining wins?

22 variants, six paradigms, one frozen protocol, ranked on the same test set.

Language supervision alone does not see Mars

34.8%

CLIP zero-shot, no
labelled Mars data

54.4%

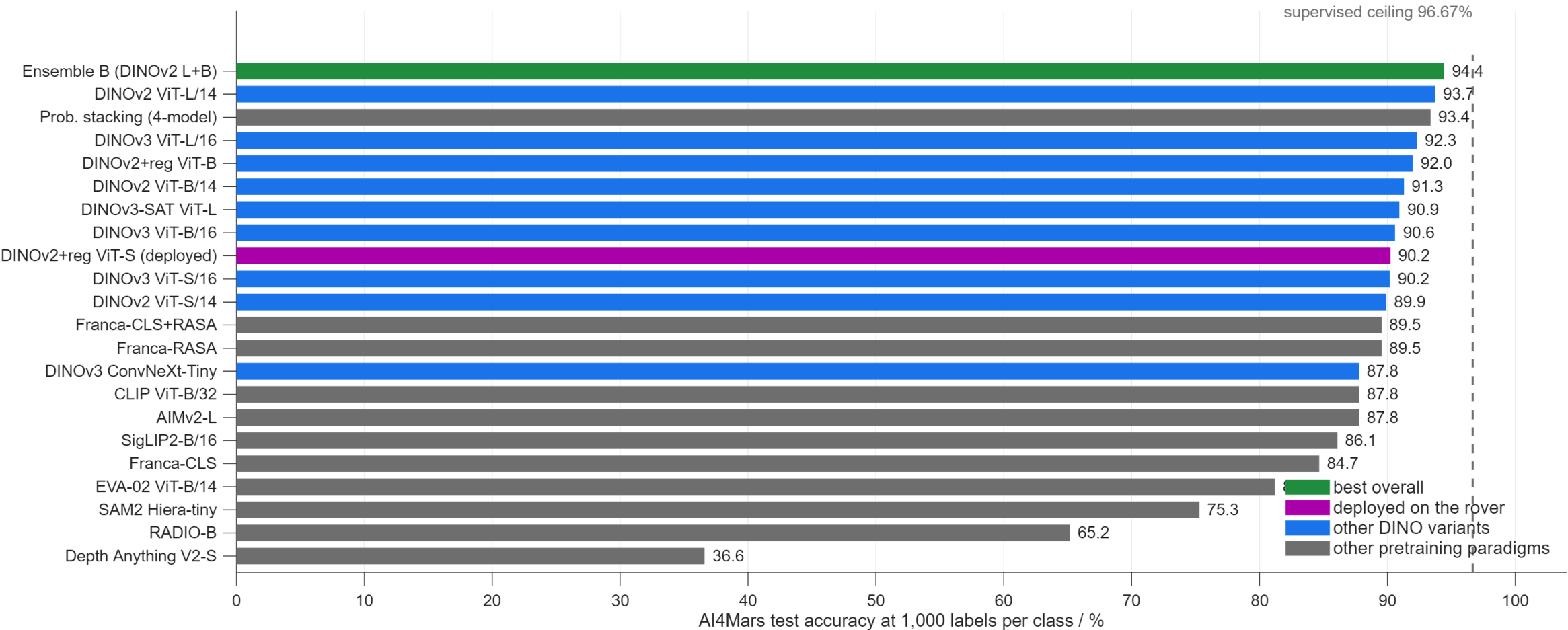
CLIP after prompt
engineering, still zero-
shot

96.67%

supervised ceiling,
trained on the full labelled
set

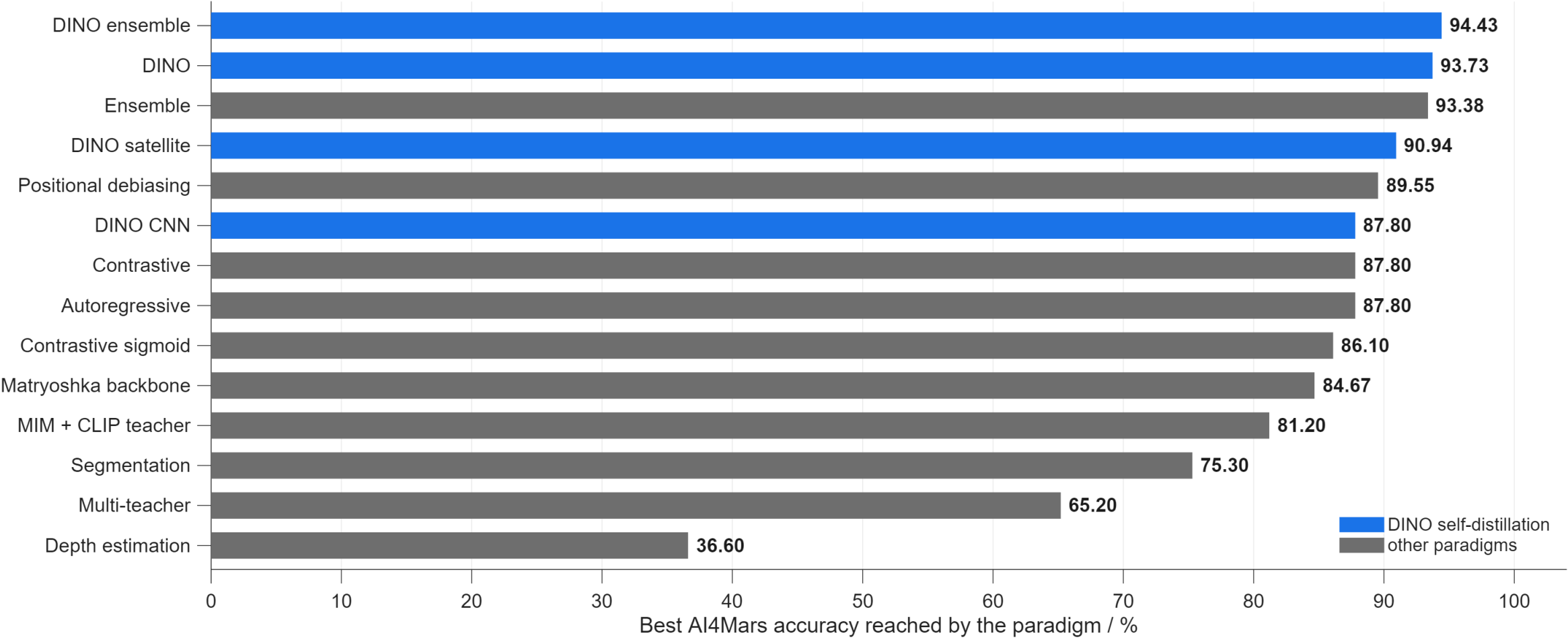
Prompt engineering nearly doubles CLIP and still leaves 42 points to the ceiling.
The wording was never the problem.

The eleven best variants are all DINO



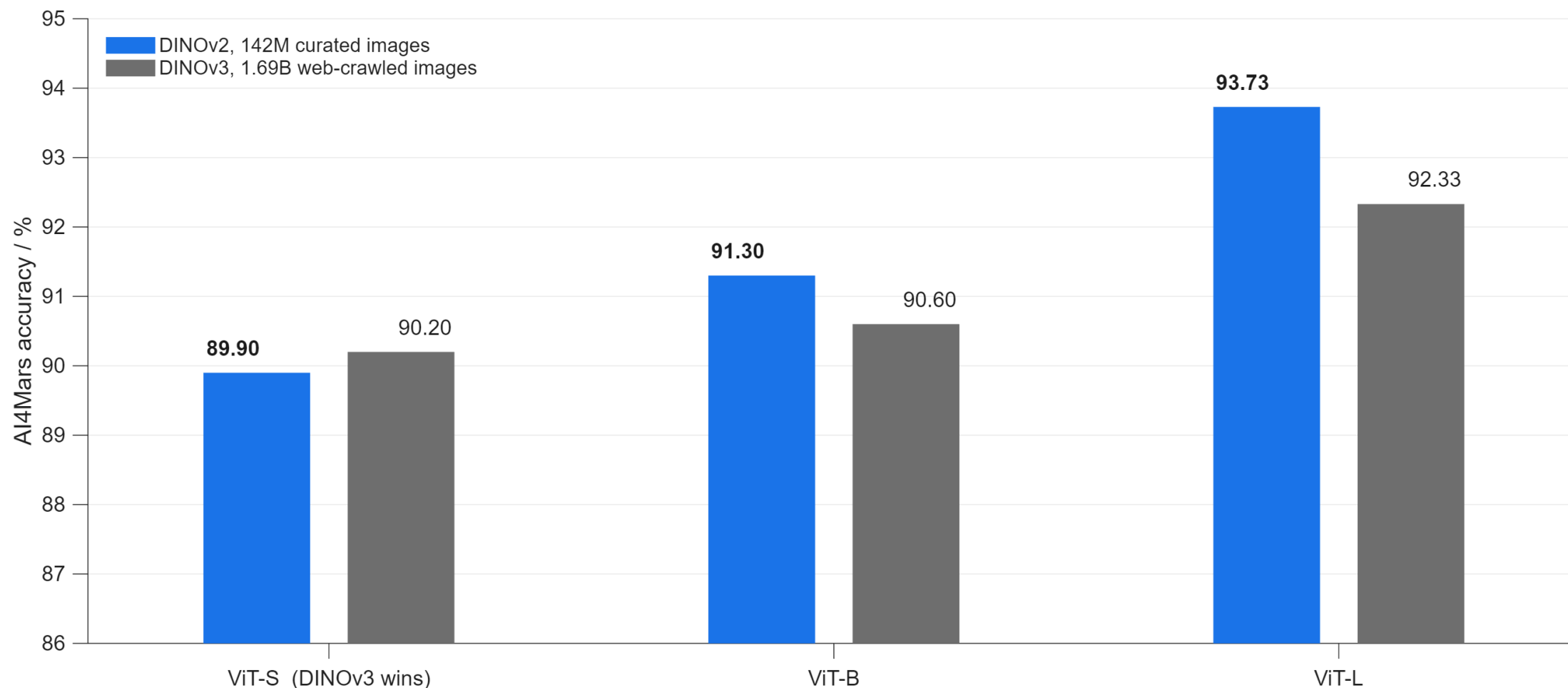
1,000 labels per class. Frozen encoders throughout.

Self-distillation beats every other paradigm



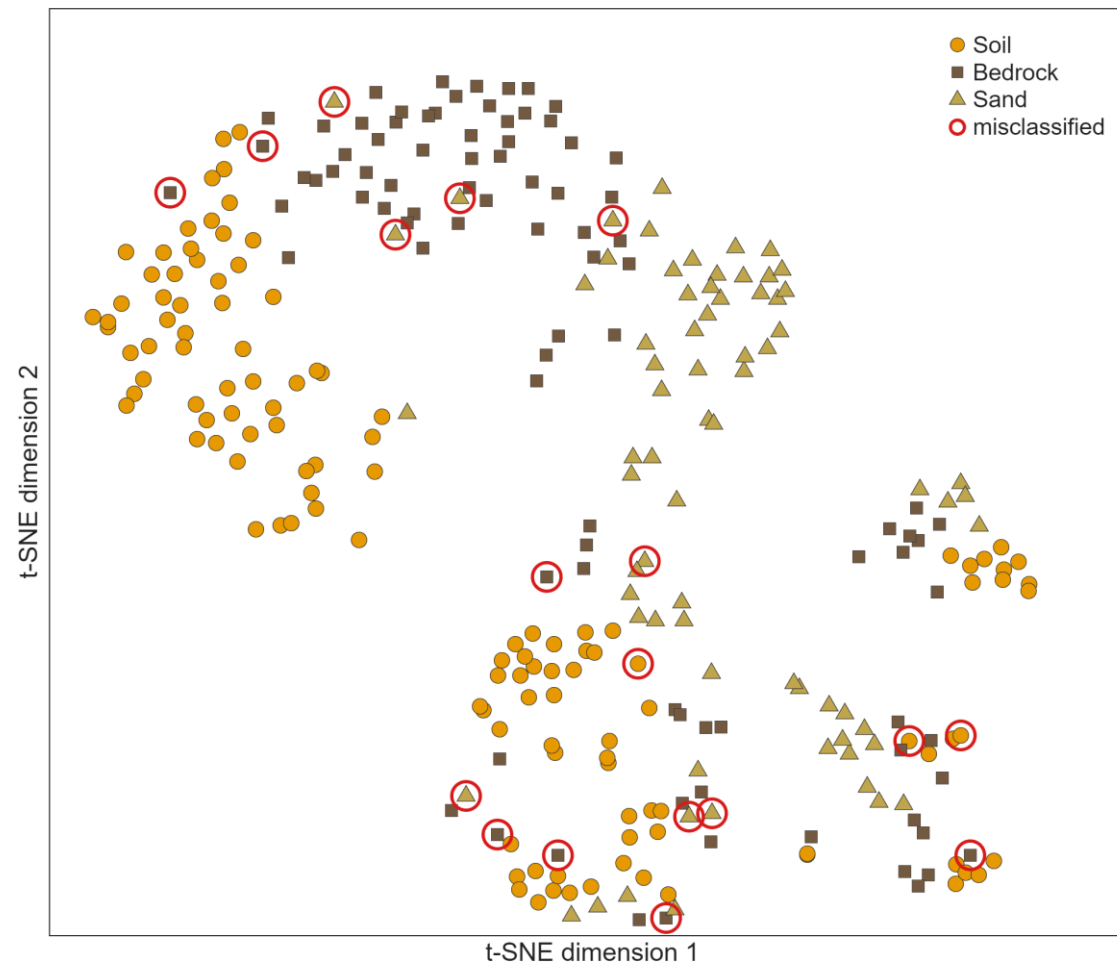
Best result per pretraining paradigm. The rival ceiling is 89.55%.

Curated data beats more data



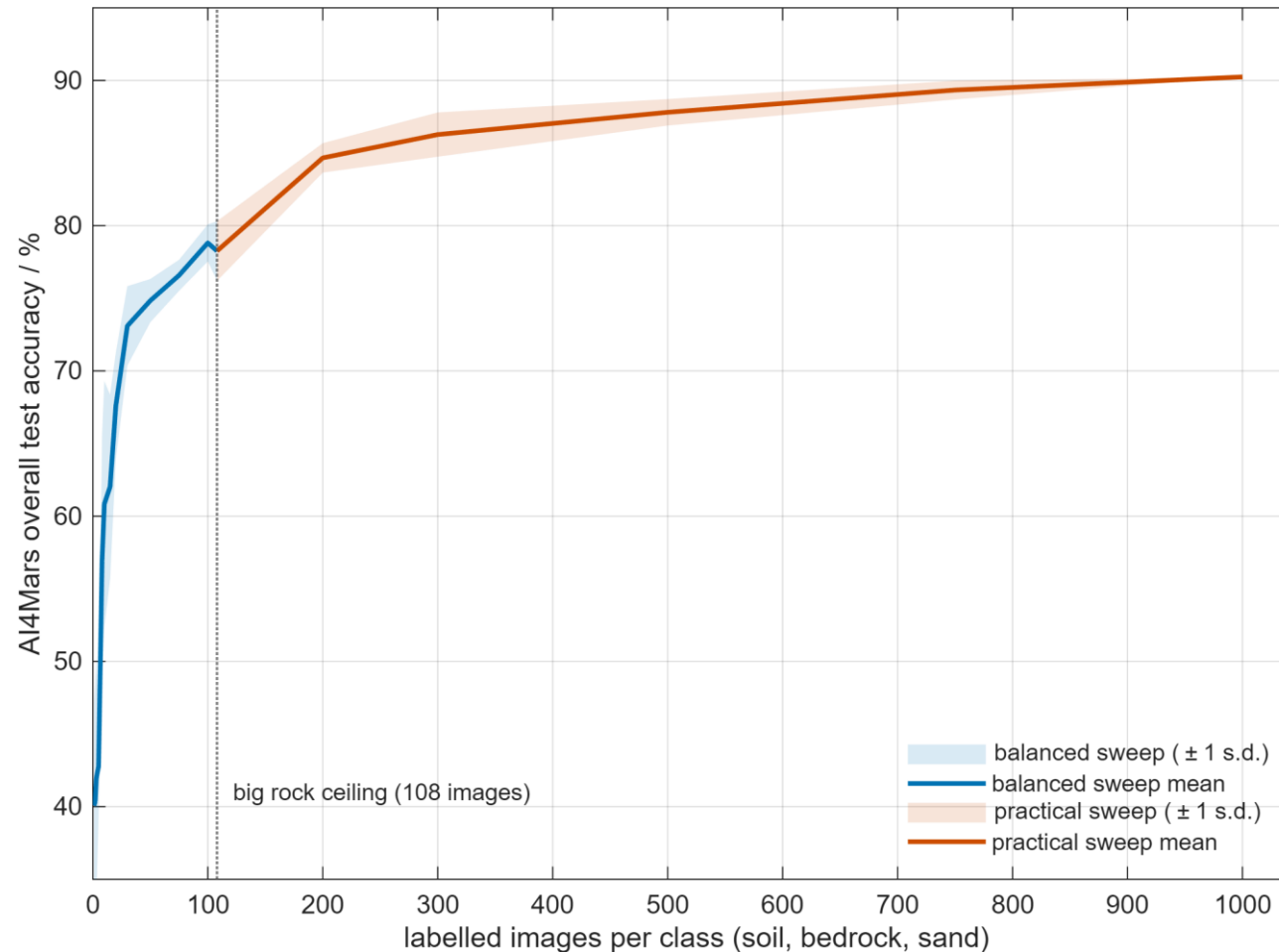
DINOv2's 142M curated images beat DINOv3's 1.69B web-crawled images at ViT-B, 91.30 against 90.60, and at ViT-L, 93.73 against 92.33.

Why: view-invariant texture, not semantic category



t-SNE of frozen DINOv2 ViT-L features on the gold test set.

Ten labels per class already gives 60.8%



Balanced training sets. Accuracy against labels per class.

The full ranking, all 22 variants

Rank	Model	Paradigm	%
1	Ensemble B (DINOv2 L+B)	DINO ensemble	94.43
2	DINOv2 ViT-L/14	DINO	93.73
3	Prob. stacking (4-model)	Ensemble	93.38
4	DINOv3 ViT-L/16	DINO	92.33
5	DINOv2+reg ViT-B	DINO	91.99
6	DINOv2 ViT-B/14	DINO	91.30
7	DINOv3-SAT ViT-L	DINO satellite	90.94
8	DINOv3 ViT-B/16	DINO	90.60
9	DINOv2+reg ViT-S (deployed)	DINO	90.24
10	DINOv3 ViT-S/16	DINO	90.20
11	DINOv2 ViT-S/14	DINO	89.90

Rank	Model	Paradigm	%
12	Franca-RASA	Positional debiasing	89.55
13	Franca-CLS+RASA	Positional debiasing	89.55
14	AIMv2-L	Autoregressive	87.80
15	CLIP ViT-B/32	Contrastive	87.80
16	DINOv3 ConvNeXt-Tiny	DINO CNN	87.80
17	SigLIP2-B/16	Contrastive sigmoid	86.10
18	Franca-CLS	Matryoshka backbone	84.67
19	EVA-02 ViT-B/14	MIM + CLIP teacher	81.20
20	SAM2 Hiera-tiny	Segmentation	75.30
21	RADIO-B	Multi-teacher	65.20
22	Depth Anything V2-S	Depth estimation	36.60

Ranks 1 to 11 are DINO or DINO ensembles without exception. Rank 12 is the best of anything else.



QUESTION 2

Does it fit on the rover?

A Raspberry Pi 4, no GPU, and a model that has to answer while the wheels are turning.

The platform: ExoMy on a Raspberry Pi 4, no GPU

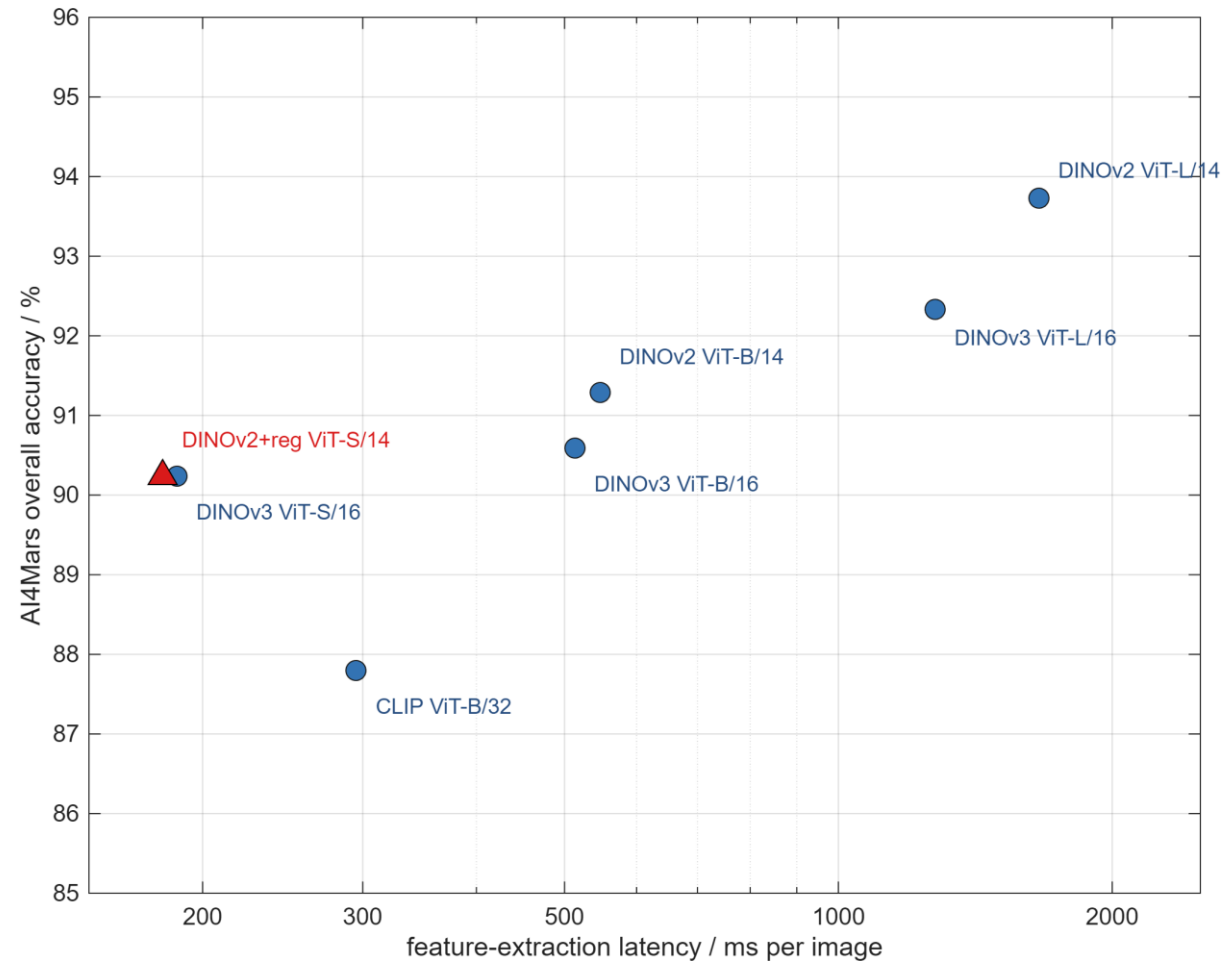


Open-source ESA ExoMy chassis, six wheels, six steering servos.

One RGB camera, one single-plane 2D LiDAR, one IMU.

Raspberry Pi 4, 8 GB. No GPU, no external compute, no network.

3.5 points of accuracy buys 14 times the model



DINOv2 ViT-L 93.73% against the deployed ViT-S 90.24%.

570 ms per image, 258 MB, on the real device

570 ms

per image, ONNX
Runtime on a Raspberry
Pi 4

258 MB

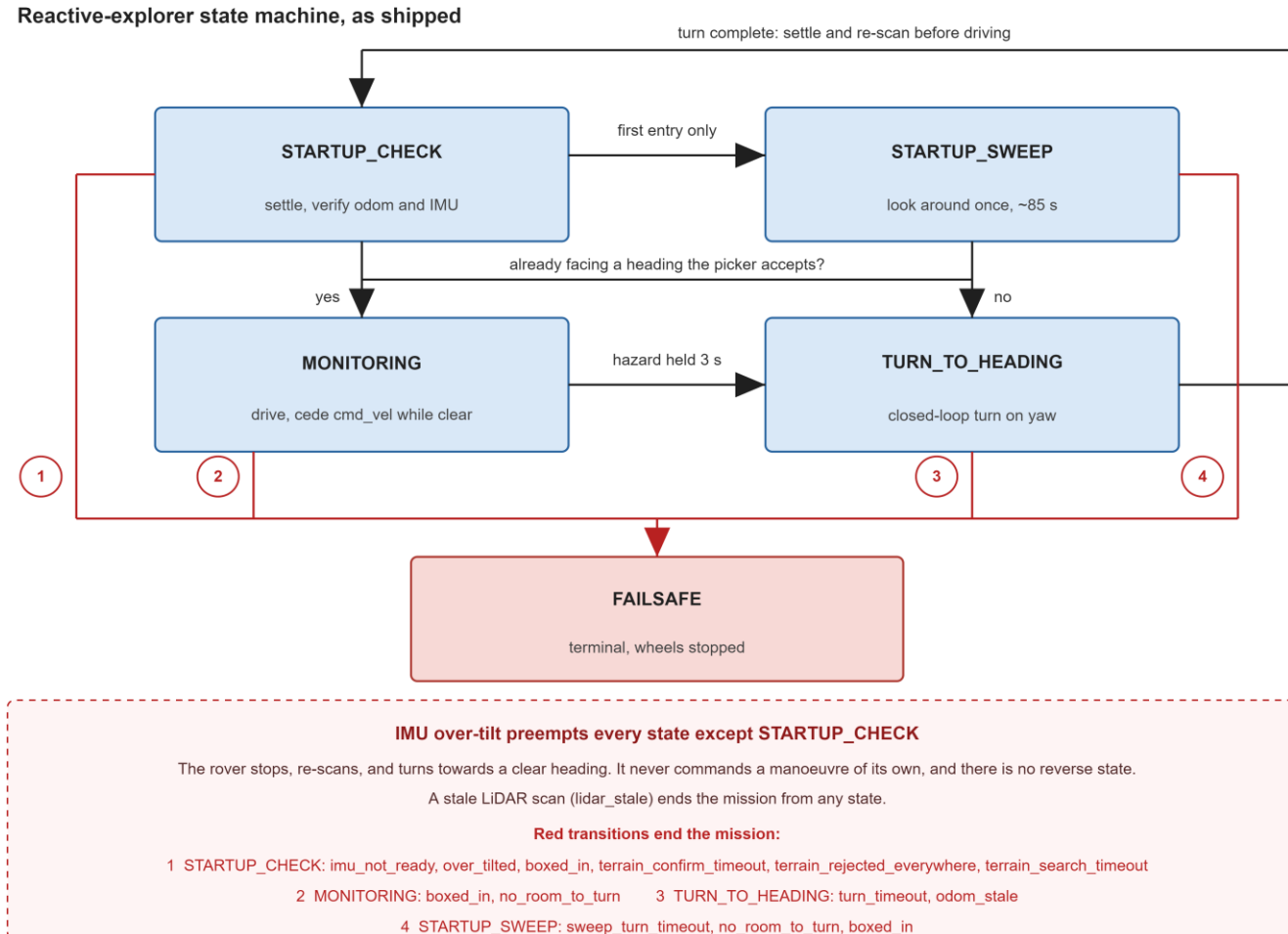
peak RAM, leaving
headroom on 8 GB

90.24%

AI4Mars accuracy of the
deployed probe

Measured on the device, not estimated from a desktop.

LiDAR and IMU are vetoes. They never name terrain



The shipped state machine. There is no reverse gear.

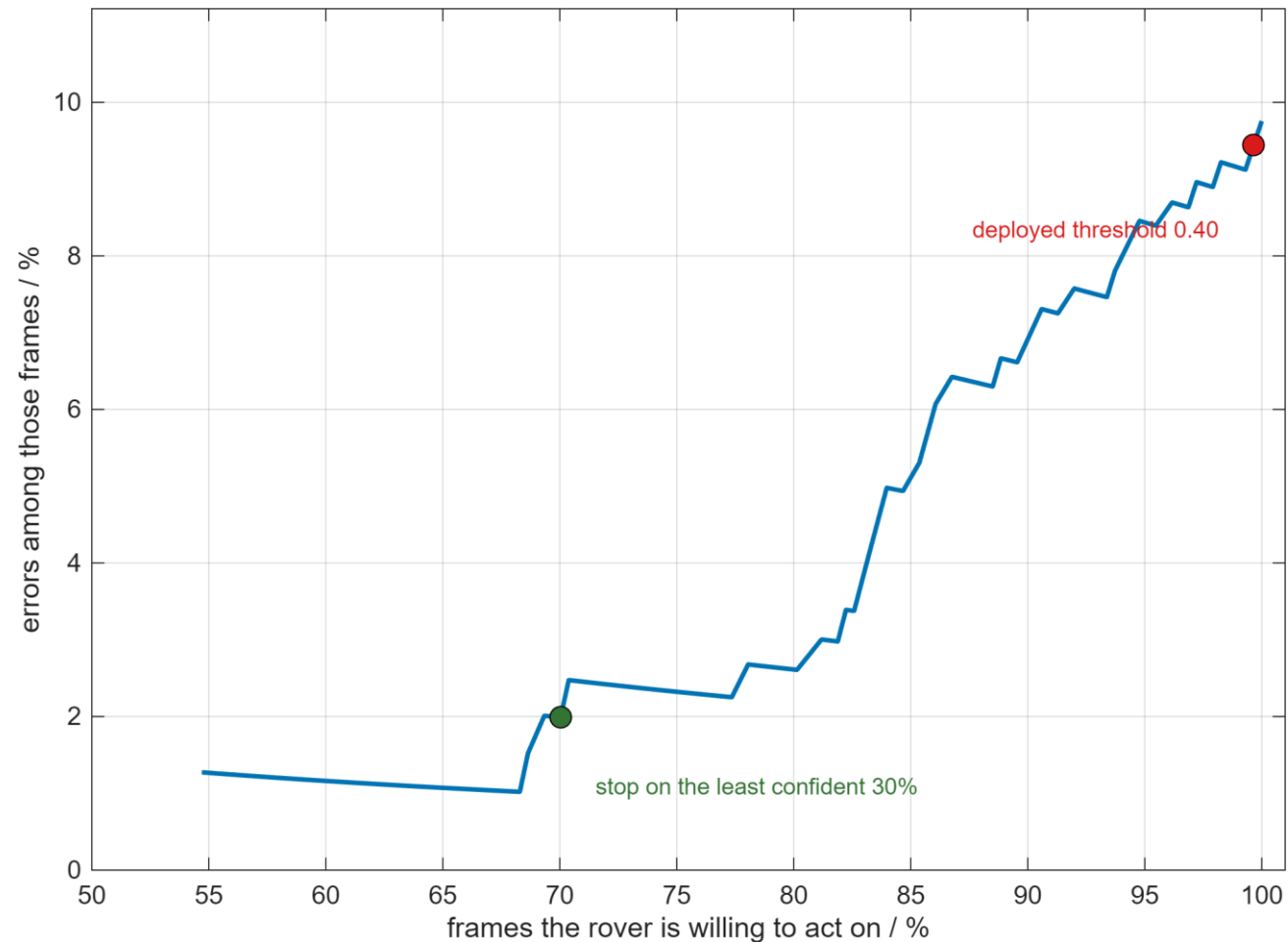


QUESTION 3

Is it safe when it is wrong?

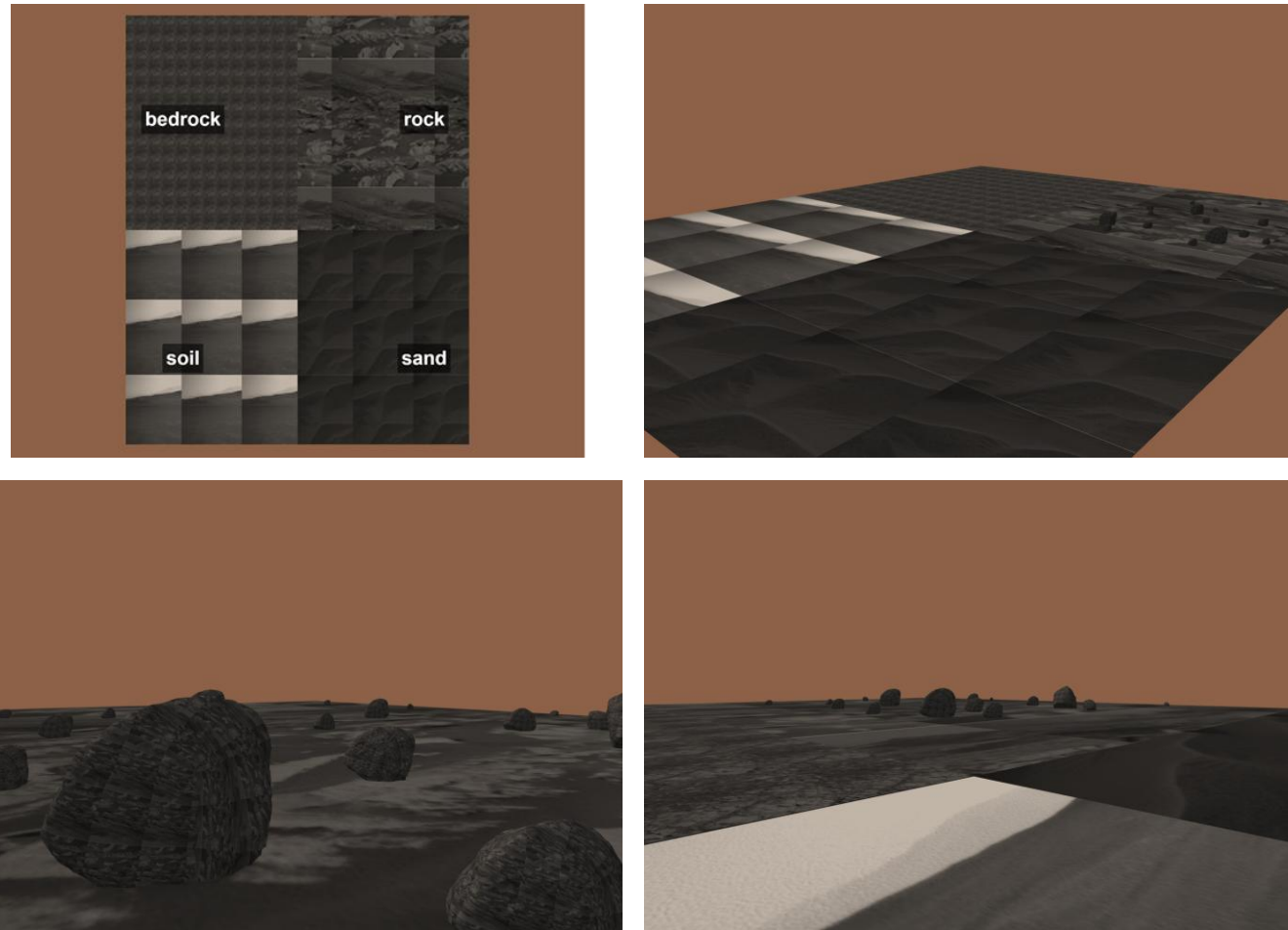
Safety does not come from being right. It comes from knowing when you are not.

A calibrated threshold turns error into delay



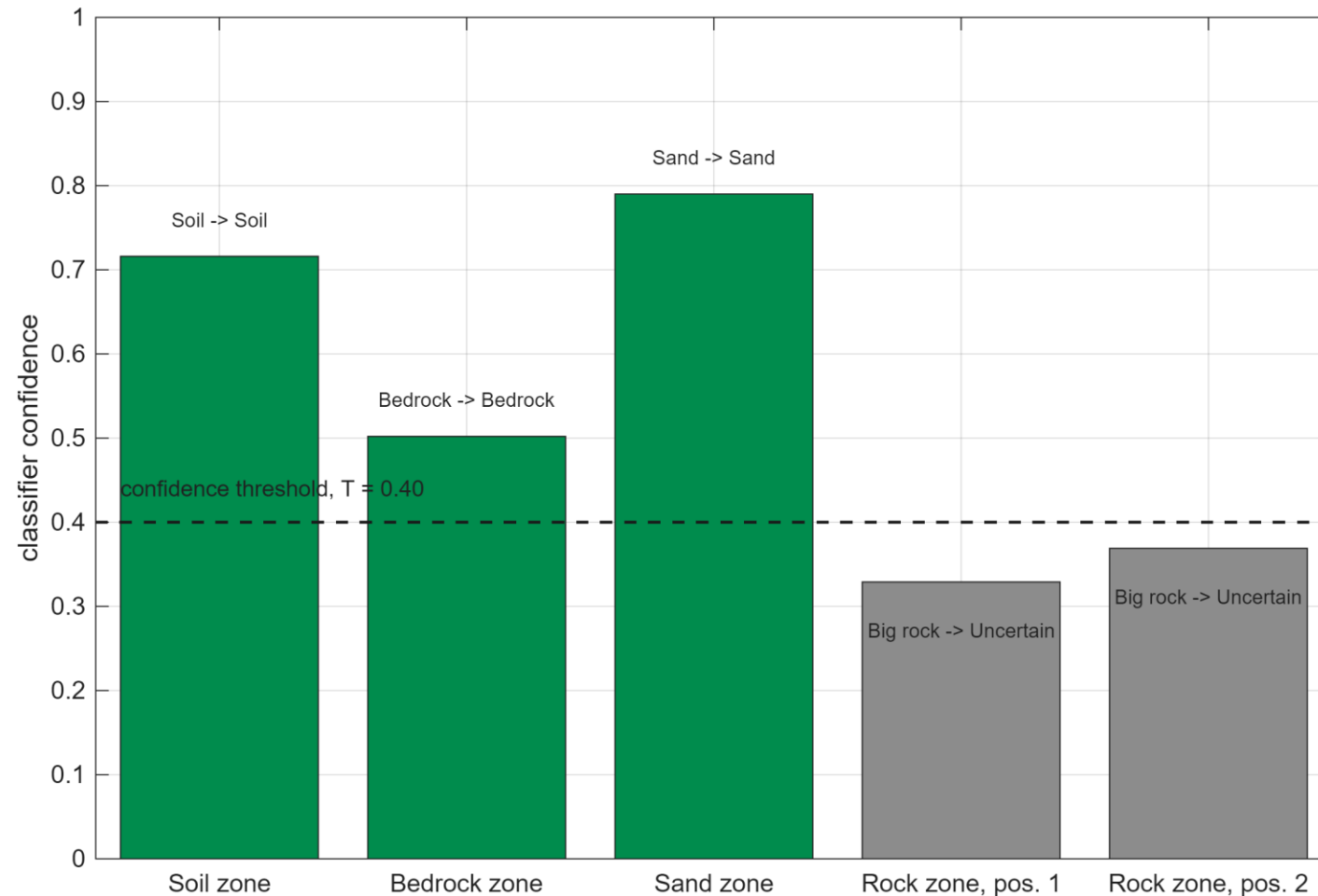
Temperature scaling cuts Expected Calibration Error by 80.8%, to 0.0325.

A benchmark world where the right answer is known



A 30 x 24 m Gazebo world with four AI4Mars-textured quadrants.

Right 3 times out of 5, safe 5 times out of 5



Terrain named correctly at 3 of 5 positions. Every one of the 5 ended safely.

Ten of ten safe endings on real hardware



Ten autonomous runs in a sandpit with a printed Mars backdrop.

Every run ended in a designed failsafe, none in a collision.

No run was ended by a terrain misjudgement.

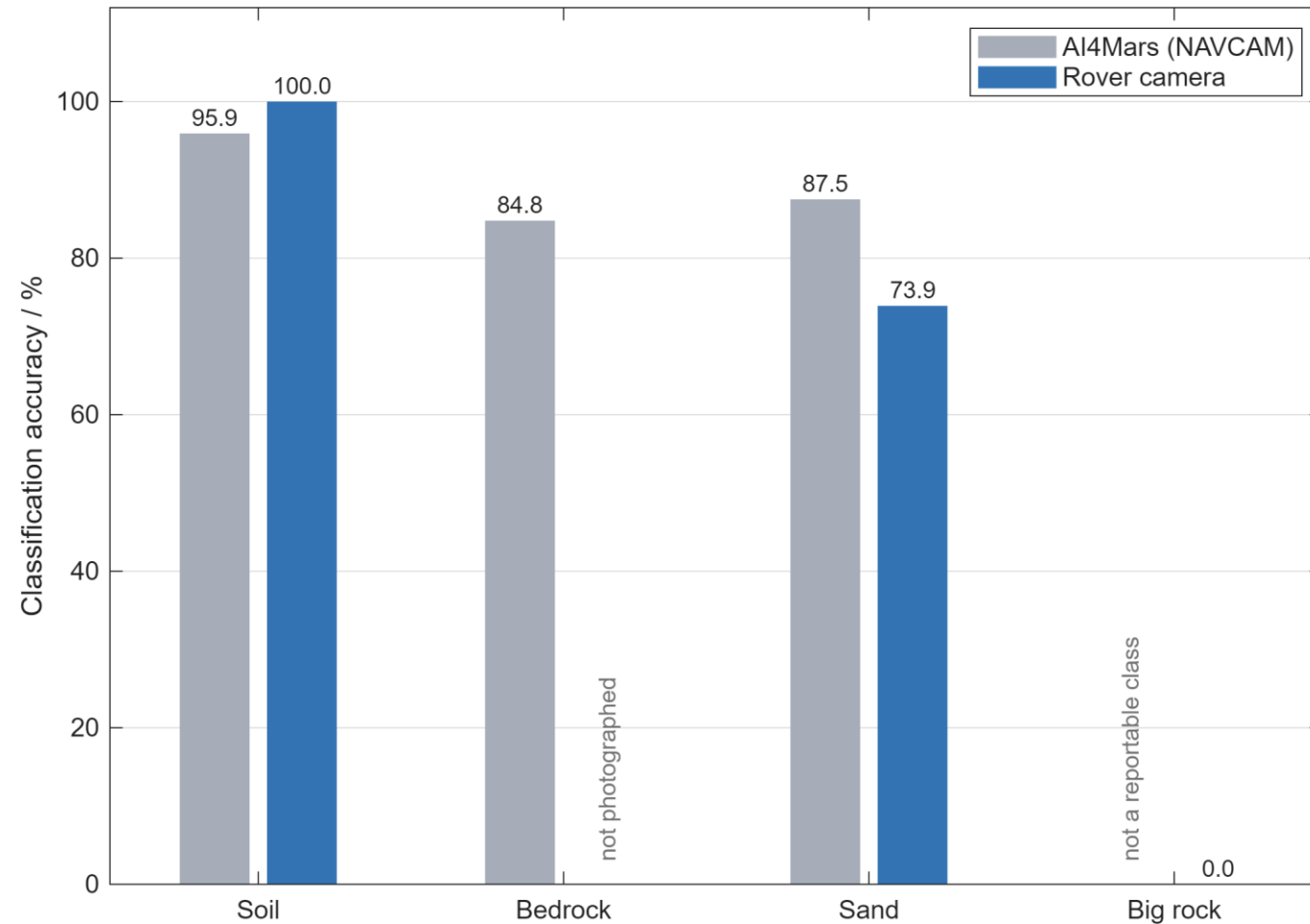


QUESTION 4

Where does it break?

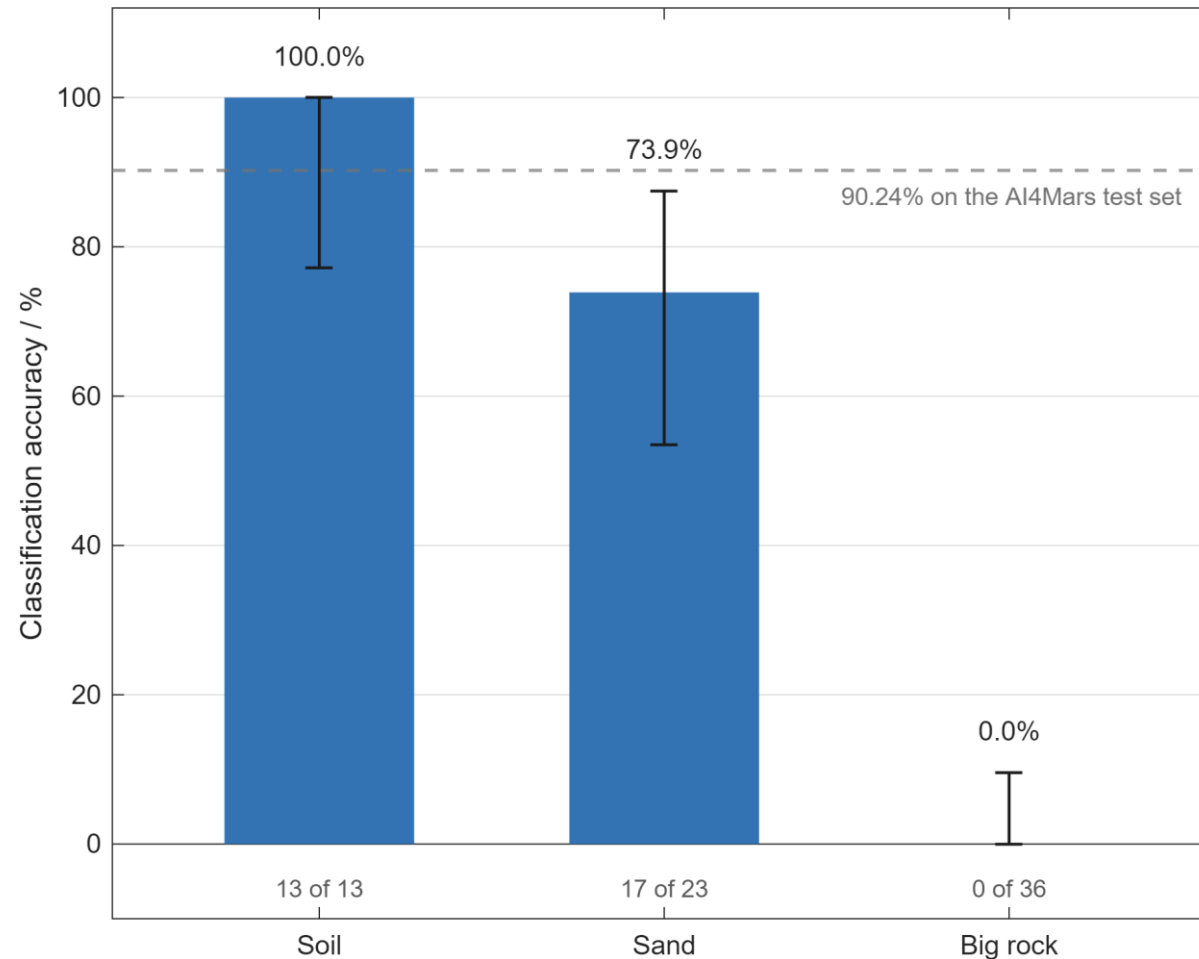
One class fails, the sensor has two blind spots, and the thesis says so.

Simulation did not predict the camera



The same model, the same classes, simulation against the rover's own camera.

Transfer is class-dependent, not uniformly poor



Real camera, 200 photographs. Soil is classified perfectly; sand holds at 73.9%.

The gap is a labelling gap, not a feature gap

Four-class probe, balanced	Accuracy
Big rock	70.0%
Soil	93.3%
Bedrock	43.3%
Sand	73.3%
Same class at 3.5 to 1 imbalance	0.0%

Big rock is outside the deployed three-class label set, so all 36 photographs are answered as soil or sand at high confidence and the 0.40 stop threshold never fires.

What the safety guarantee does and does not cover

1

It holds in simulation, where every failure mode was traced to a cause.

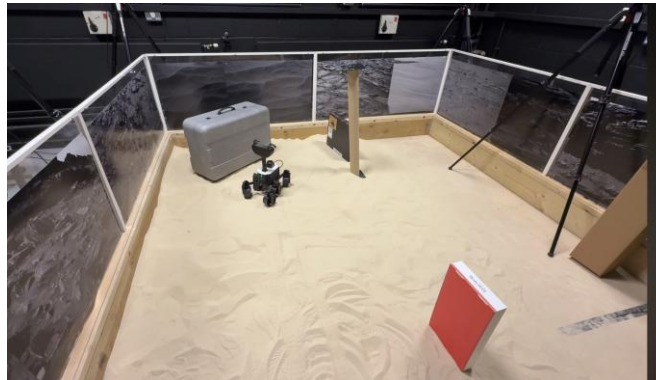
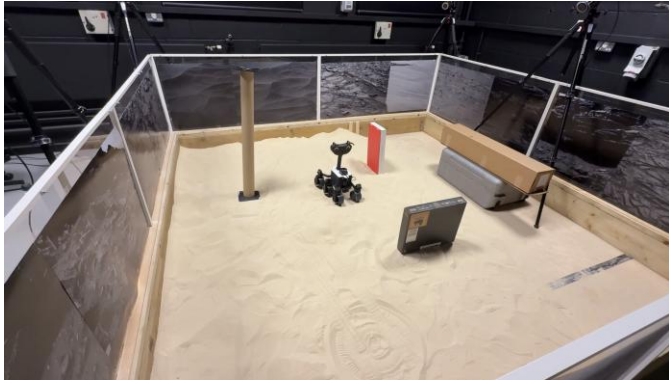
2

It holds on hardware for the classes the probe can name.

3

It does not hold for a hazard outside the label set, because the stop never fires.

A single-plane LiDAR has two different blind spots



On a slope, the scan plane tilts with the chassis and stops tracking the obstacle ahead.

On flat ground, anything below the fixed scan height is never seen. A chair's splayed legs were not.

Separate mechanisms with the same symptom, and they must not be folded into one.

Every hardware run that ended early ended on geometry, not on a terrain misjudgement.



PART 3

Conclusions

What frozen features were enough for, and what they were not.

Frozen features are enough to start, not enough to finish

94.43%

best frozen-encoder
accuracy, 2.24 points
from the supervised
ceiling

5 of 5

safe outcomes in Gazebo
despite 3 of 5 terrain
accuracy

10 of 10

safe endings on the real
rover

No gradient update to any encoder, at any point in this work.

Where the remaining work belongs

1

Collect real rover imagery for big rock and return the class to the label set.

2

Replace the single-plane LiDAR, which has two distinct blind-spot mechanisms.

3

Test in-field adaptation, where the probe is refitted from a few labels on site.

Six contributions

- 1 A prompt engineering study for Mars terrain with CLIP, 34.8% to 54.4% with a hybrid ensemble prompt.
- 2 The most complete frozen-encoder comparison on AI4Mars to date, 22 variants across six families.
- 3 The best reported frozen-encoder accuracy on AI4Mars at 94.43%, with no GPU fine-tuning.

Three more follow: the ROS2 package, the Raspberry Pi benchmark, and the calibrated safety policy.

Thank you

1

Code: github.com/thupthaiNK/Foundational-Model-for-Rover-Autonomy

2

Video: eight autonomous hardware runs on YouTube

3

Project page and poster: see the QR code

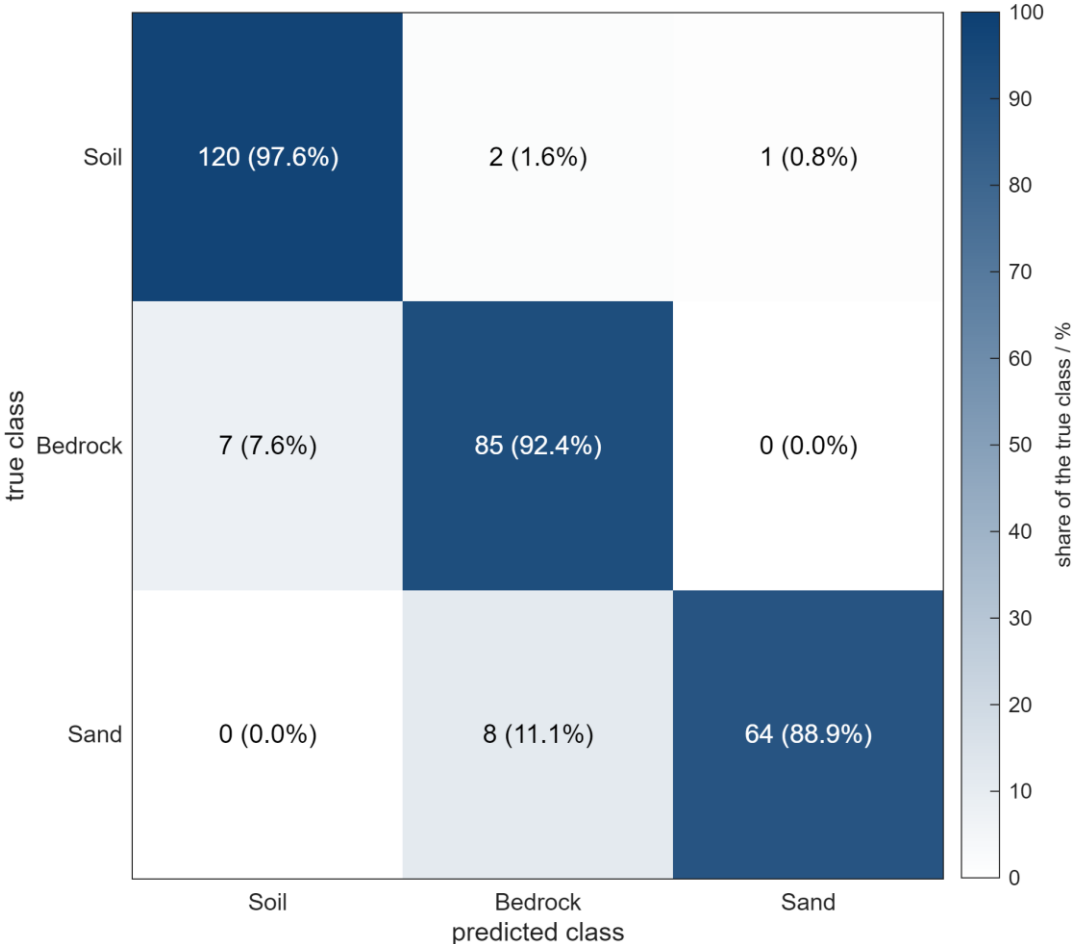


APPENDIX

Supporting detail

The tables and checks behind the claims, for questions that go deeper.

Errors concentrate on bedrock, not on the weakest class



DINOv2 ViT-L, 93.73%. Every class clears 88%. The errors are not scattered: bedrock takes all 8 of sand's and 2 of soil's 3, while soil and sand are mistaken for each other once in 195 images.

The ensemble is not significantly better than one encoder

94.43%

Ensemble B, two DINOv2 encoders

93.73%

DINOv2 ViT-L alone

$p = 0.75$

McNemar test on the 10 images where they disagree

The recommendation is therefore the single encoder wherever offline compute allows it.

Calibration, not accuracy, is what the safety policy rests on

0.0325

Expected Calibration
Error after temperature
scaling

80.8%

reduction from the
uncalibrated model

0.461

the fitted temperature

A stated confidence of 70% now means right about 70% of the time,
which is what makes a threshold meaningful.

What was tested, and what testing cannot cover

- 1 347 of the 652 unit tests across the three ROS2 packages exercise the behaviours claimed here.
- 2 Reactive exploration 147, stuck detection 80, path planning 52, frontier 26, fusion 15.
- 3 Tests pin behaviour against regressions. They do not establish that the behaviour is right in the field.

Both ROS2 setups must be sourced to reproduce the count, or 16 files fail to import and it reads 241.